

Organization of a Computer — Detailed Lecture Notes

JKSSB Computer Awareness

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Video lesson: [How Computers Actually Work: CPU, Memory & I/O Explained](#)

04.1 The functional units of a computer

Start with the office desktop. A clerk types marks on the keyboard, the CPU processes them through a spreadsheet formula, the monitor and printer present the result, and the file is kept on the SSD for later use. This single machine contains every functional unit that the exam asks you to name.

Definition (Functional unit). *A major part of a computer that performs one broad job — accepting input, producing output, holding data, performing operations, or controlling the machine.*

Unit	Job in this lesson
Input unit	Accepts data and instructions from the outside world.
Output unit	Presents the result to the user.
Memory (primary storage)	Holds data and instructions currently in use.
Arithmetic Logic Unit (ALU)	Performs arithmetic and logical operations.
Control unit	Directs the sequence of operations and the flow of data.

The memory unit described here is *primary storage*, that is, main memory. Secondary storage such as a hard disk or an SSD is a separate idea and is treated in Section 4.

04.2 The CPU: ALU, control unit and registers

The **Central Processing Unit (CPU)** is the processing centre of the computer. It is built from three parts: the arithmetic logic unit, the control unit, and registers.

Definition (Central Processing Unit (CPU)). *The unit that fetches, decodes and executes instructions; it is formed by the ALU, the control unit and the registers.*

Definition (Arithmetic Logic Unit (ALU)). *The part of the CPU that performs arithmetic operations (such as addition and subtraction) and logical operations (such as comparison) on binary data.*

Definition (Control unit). *The part of the CPU that fetches and decodes instructions, issues control signals, and coordinates the operation of all other units.*

Definition (Register). *A small, very fast storage location inside the CPU used to hold data, addresses or control information during processing.*

The ALU does the calculating, but it does not decide what to do. The control unit decides the sequence of operations and the flow of data, and then tells the ALU and the other units when to act. Registers are the CPU's own working storage; they are not the same as main memory or secondary storage.

Example 04.2.1. *In the marks spreadsheet, the formula that adds three marks is carried out by the ALU. The control unit has already fetched the formula's instruction from memory, decoded it, and signalled the ALU to act. The intermediate values are kept in registers while the operation runs. The final total is then written back to memory and later shown on the monitor.*

04.3 Input and output units

The input unit brings data and instructions into the system. The output unit presents the result.

Input devices	Output devices
Keyboard, mouse, scanner, microphone, light pen, digitizer	Monitor, printer, plotter, speaker, sound-card output

A device is classified by the role it plays in the described task, not by its shape. A monitor is normally an output device because it presents results. A scanner is an input device because it brings data in. When a question lists devices, mark each one as input or output before choosing the option.

04.4 Memory (primary storage) versus secondary storage

This is a high-value contrast.

Definition (Primary storage (main memory)). *The memory the CPU accesses directly while processing; it holds the programs and data currently in use. Random Access Memory (RAM) is volatile; Read Only Memory (ROM) is non-volatile.*

Definition (Secondary storage). *Storage that keeps data and files permanently but is not accessed directly by the CPU during processing; examples are the hard disk, the Solid State Drive (SSD) and the pen drive.*

The key rule for the exam is that the **CPU works with main memory, not with secondary storage**. When a program on the hard disk is run, it is first loaded into main memory; the CPU then works from main memory. Secondary storage is reached through main memory and the operating system.

Example 04.4.1. *The spreadsheet file is saved on the SSD. When the clerk opens it, the file is first loaded into RAM. All the additions and comparisons are then performed on data held in RAM and in the CPU registers. Writing the result back to the SSD happens later, through the operating system. At no point does the ALU process the file straight from the SSD.*

04.5 The system bus: address, data and control

The **system bus** is the set of wires that carries information between the CPU, memory and the other units. It has three parts, and their directions differ — this is a common trap.

Bus	Direction	Role
Address bus	CPU $\rightarrow$ memory	Carries the address of the memory location the CPU wants to use.
Data bus	Bidirectional	Carries the actual data and instructions between the CPU and memory.
Control bus	CPU $\rightarrow$ units	Carries control and timing signals, such as read and write.

Definition (System bus). *The address bus, data bus and control bus taken together; they connect the CPU with memory and the input/output units.*

Remember the three questions: the address bus answers *where*, the data bus answers *what*, and the control bus answers *when and how*. The address bus is one-way (from the CPU outwards) and carries the address, not the contents of the location. The data bus is two-way. The control bus carries signals, not data or addresses.

Example 04.5.1. *The CPU wants the value stored at memory location 200. It places the address 200 on the address bus and sends a read signal on the control bus. The memory puts the contents of location 200 onto the data bus, and the CPU receives it. The address travelled out on the address bus; the data travelled back on the data bus.*

04.6 The motherboard, expansion slots and interfaces

The **motherboard** is the main circuit board that connects and links the CPU, main memory and the buses. Expansion slots on the motherboard allow extra cards to be added, such as a graphics card or a network card. Interfaces and ports connect external devices — keyboard, printer, monitor — to the system. The system bus runs along the motherboard between the CPU and the other units, so the motherboard is the physical platform on which the units communicate.

04.7 The stored-program (von Neumann) idea

In early machines, changing the task often meant rewiring the machine. The **stored-program concept** changed this.

Definition (Stored-program concept (von Neumann architecture)). *The design in which a program's instructions are stored in main memory together with the data, so the CPU can fetch and execute them without rewiring the machine.*

Because instructions and data share the same memory, the CPU fetches an instruction from memory, executes it, and moves on to the next. This repeating pattern is the instruction cycle, described next.

04.8 The instruction cycle

The **instruction cycle** is the sequence the CPU follows for each instruction.

1. **Fetch**: the CPU obtains the next instruction from main memory.
2. **Decode**: the control unit works out what the instruction means.

3. **Execute**: the ALU or another unit performs the operation.
4. **Store** (when required): the result is written back to memory or to a register.

The order is fixed: fetch, then decode, then execute. The optional store step writes the result back. Data and instructions move between the units over the system bus, and the whole cycle repeats for the next instruction.

Example 04.8.1. *Consider an instruction “add the value in memory to a register”. During fetch, the CPU gets the instruction from memory. During decode, the control unit identifies it as an addition and signals the ALU. During execute, the ALU performs the addition. The updated value is then stored back in the register or in memory, and the cycle begins again.*

Three distinctions prevent most errors in this topic:

1. The CPU accesses main memory directly, not secondary storage.
2. Address, data and control buses are not interchangeable: the address bus is one-way and carries addresses; the data bus is two-way and carries data and instructions; the control bus carries control signals.
3. Memory (primary) holds current work; storage (secondary) keeps files permanently.

Exercises

1. Name the five functional units of a computer and state one job of each.
2. What are the three parts of the CPU, and what does each do?
3. State the direction and role of the address bus, the data bus and the control bus.
4. A file is opened from a hard disk and edited. Explain why the CPU does not process the file directly from the hard disk.
5. List the steps of the instruction cycle in order.

Solutions

1. Input unit (accepts data and instructions); output unit (presents results); memory/primary storage (holds data and instructions in use); ALU (performs arithmetic and logical operations); control unit (directs operations and data flow).
2. The ALU, the control unit and the registers. The ALU computes; the control unit fetches, decodes and signals; the registers hold data and addresses during processing.
3. Address bus: one-way from CPU to memory, carrying the address of the location. Data bus: bidirectional, carrying data and instructions. Control bus: carries control and timing signals from the CPU to other units.
4. The CPU accesses only main memory directly. The file is first loaded into main memory, the CPU processes the copy in main memory and registers, and the operating system writes the changes back to the hard disk.
5. Fetch, decode, execute (and store when a result must be written back).